

Radiological Parameters Evaluation as Risk Factors of Proximal Junctional Kyphosis following Long Segment Fusion in Adult Spinal Deformity Surgery: A Literature Review

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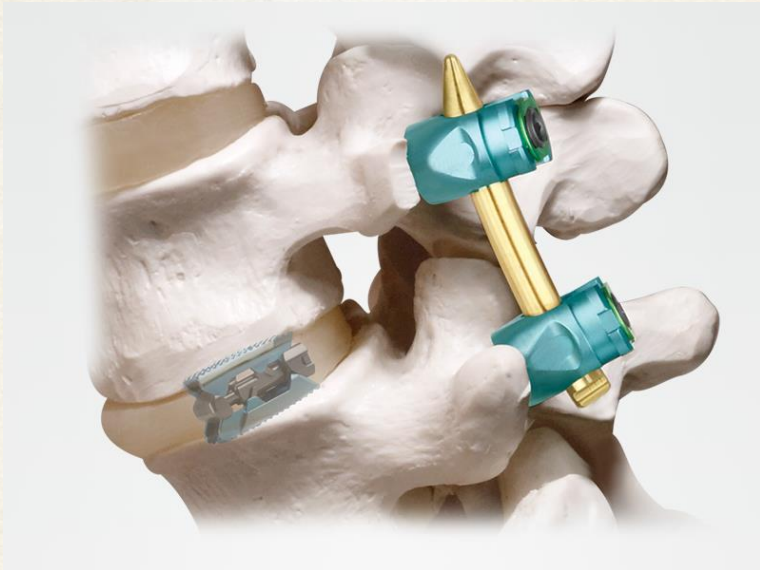
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Introduction

The prevalence of adult (de novo) scoliosis and adult spinal deformities (ASD) **has risen**

➔ Impact in quality of life



➔ resulted in **proximal junctional kyphosis (PJK)**

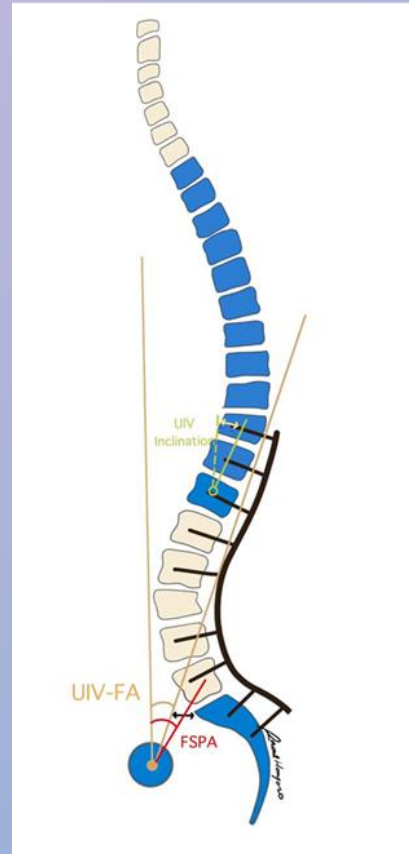
long segment stabilization, advanced technology, and various types of implants

Decision on Surgery based on:

Degenerative process

Surgical planning

STUDY AIM:



Radiological parameters that influence
PJK in ADS corrective deformity surgery
and prevention techniques

Materials and Methods

- PubMed papers from January 2001 - June 2024
- MeSH terms "proximal junctional kyphosis", "adult spinal deformity", and "risk factor"

Search Strategy

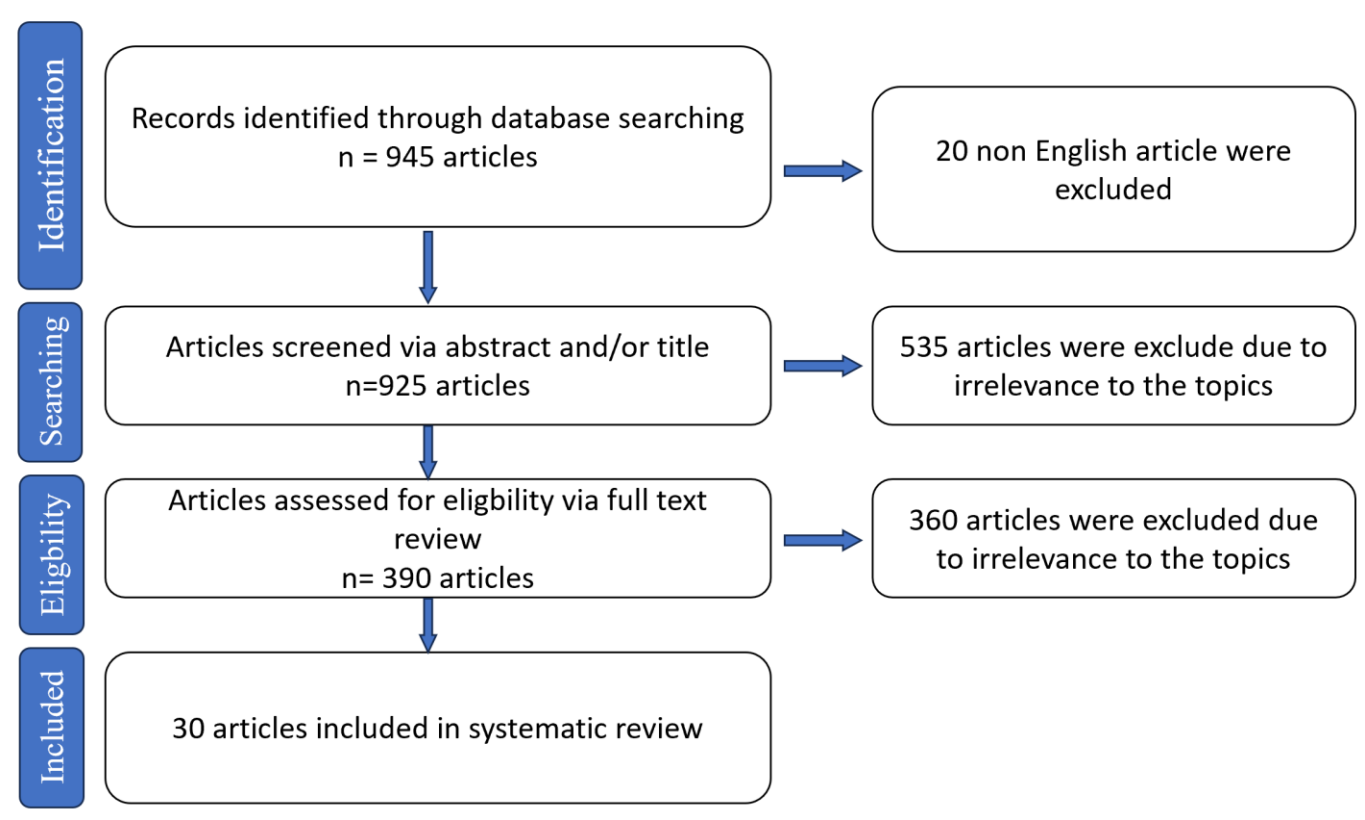
Selection Criteria

- Peer-reviewed studies in English
- Radiology parameters associated with incidence PJK In patients with ASD who had correction deformity and instrumented fusion
- Cohort, clinical trials, case-control, or cross-sectional studies
- Full papers were available to access

- Screened and evaluated abstracts based on 2020 **PRISMA guidelines**
- All data were presented as narratives

Data Extraction

PRISMA Flow diagram

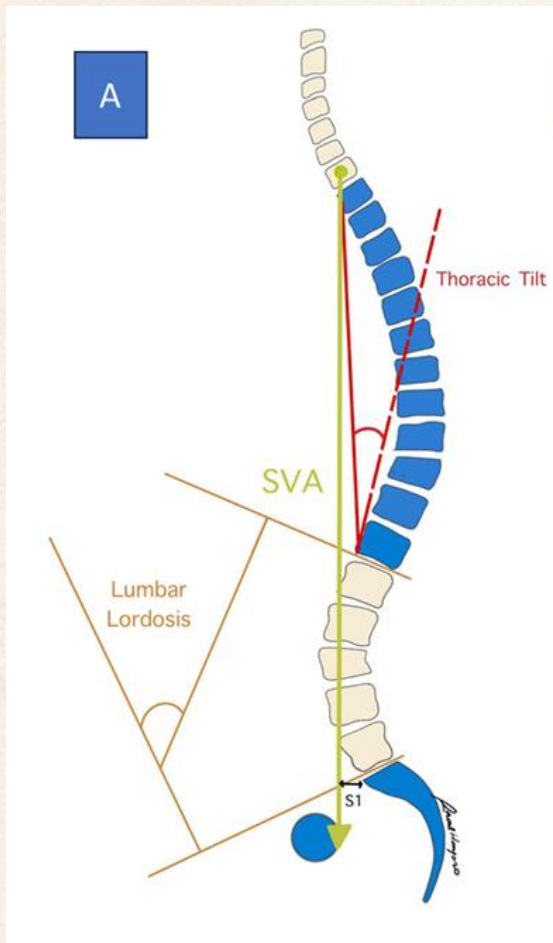


Thoracal Tilt (TT) as a Predictor of PJK

Angle between T1 superior end plate and T12 inferior endplate.

- Positive: T1 anterior to T12
- Negative: T1 posterior to T12.
- Higher TT indicates potential thoracic kyphosis imbalance
- Rebalancing the TT in a sagittal neutral position

Qiu W, Zhou S, Han G, Chen Z, Chen Z, Ding L, et al. The Role of Thoracic Tilt in Identifying Thoracic Compensation and Predicting Proximal Junctional Kyphosis in Degenerative Lumbar Scoliosis. Spine (Phila Pa 1976). 2023;48(20):1446-54.



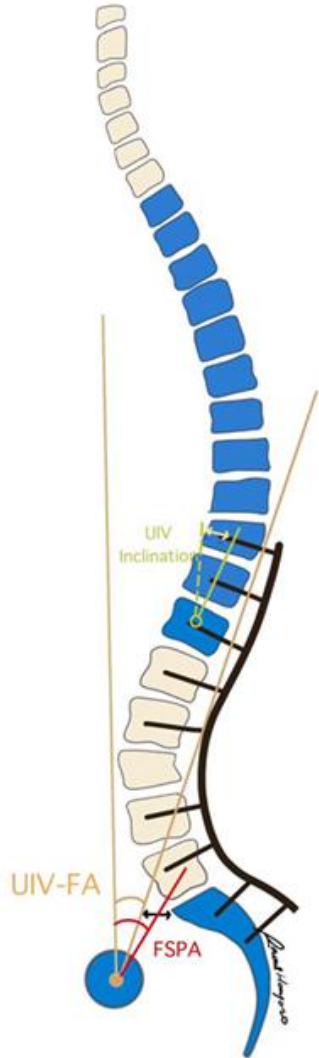
Fused Spinopelvic Angle (FSPA)

- Lower FSPA increases PJK risk.
- Increase the FSPA by adjusting the PI-LL and LDI during a spine reconstruction surgery for ASD in order to prevent PJK.

UIV Inclination (UIV)

The angle between the vertical and the best-fit line crosses the center of the UIV, UIV-1, and UIV-2

Make the alignment on the 3 upper level of fusion approaching the virtual vertical line.



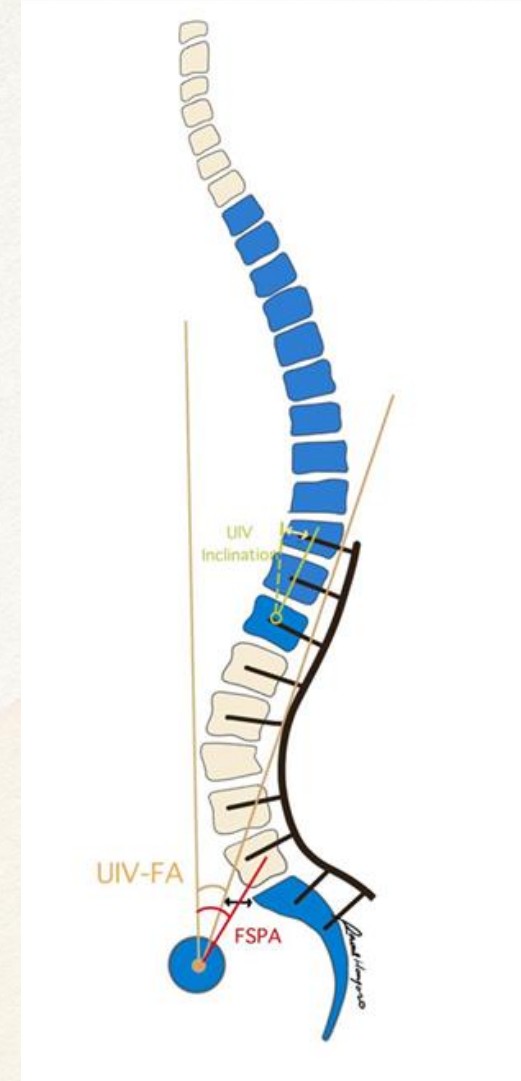
RESULTS

sagittal vertical axis (SVA)

- Higher preoperative SVA in PJK patients compared to non-PJK patients.
- Overcorrection linked to increased SVA leading to higher PJK risk
- Proposed to aim lower SVA post-operative

Upper Instrumented Vertebrae - Femoral Angle (UIV-FA)

- The angle subtended by a line from the UIV center to the center of the femoral heads to a vertical line.
- Position the UIV center as close as possible to the center of femoral head (<5 degree is more preferable)



Park SJ, Kim HJ, Lee CS, Park JS, Jung CW, Lee JS, et al. Clinical Significance of Lordosis Orientation on Proximal Junctional Kyphosis Development in Long-Segment Fusion Surgery for Adult Spinal Deformity. World Neurosurg. 2024;183:e282-e92.

Katsuura Y, Lafage R, Kim HJ, Smith JS, Line B, Shaffrey C, et al. Alignment Targets, Curve Proportion and Mechanical Loading: Preliminary Analysis of an Ideal Shape Toward Reducing Proximal Junctional Kyphosis. Global Spine J. 2022;12(6):1165-74.

Line BG, Bess S, Lafage R, Lafage V, Schwab F, Ames C, et al. Effective Prevention of Proximal Junctional Failure in Adult Spinal Deformity Surgery Requires a Combination of Surgical Implant Prophylaxis and Avoidance of Sagittal Alignment Overcorrection. Spine (Phila 1976). 2020;45(4):258-67.

RESULTS

INDONESIA

Importance of Aligning Correction with Patient Age

- No definitive "one-size-fits-all" correction target.
- Consider degeneration variances among individuals

Recommendation by Roussouly et al.

- Preserve physiological alignment to prevent further deformity

Prasse et al.'s Approach

- Anatomical reconstruction based on Roussouly types
- Formula: $L4S1 = 0.66 \times L1S1$.
- Adjustments must be tailored to individual patient conditions and goals.

Conclusion

Degeneration impacts spinal alignment and surgical outcomes.

Individualized approaches are crucial to mitigate PJK risks.

Ongoing research needed to refine surgical planning and outcomes.

